

Darshan Patel

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SUMMARY

Machine learning engineer with 9+ years building fraud, abuse and account-security systems. Currently leading Account Takeover (ATO) risk control for TikTok US across **150mn+ DAU**. Hands-on across the full model lifecycle (sequence models, gradient-boosted trees, graph analytics and unsupervised detection) on structured and behavioral data, from feature design and label governance to low-latency production deployment and monitoring against adaptive adversaries.

EXPERIENCE

TikTok U.S. Data Security

New York, NY

Lead Machine Learning Engineer

December 2023 – Present (2 years 10 months)

- Serving as the team lead for the TikTok US's Account Security function that provides Account Takeover (ATO) risk control to **150mn+ DAU**. Led the technical design to structure all ML models and strategies into defense, perception and label governance pillars to create a feedback loop to consistently beat adversarial activity.
- Spearheaded the design, implementation, launch and monitoring of the Account Takeover (ATO) user journey sequence model to increase ATO perception from **5% to 60%**.
- Guided senior MLEs to address targeted problems of sophisticated phishing attacks with classical Catboost models with low inference latency in defense, and reduced phishing prevalence by **15% QoQ**.
- Detected, investigated and mitigated a sophisticated hacker network that performed a **surgical attack on high profile news and celebrity accounts**, reducing exposure by **95% (recall)**.
- Designed, led and rapidly executed a critical High Comprised Value Accounts segmentation project with the objective of enhancing Account Security measures to improve coverage from **1% MAU to 5% MAU**. Segmentation results are live and used across TikTok's all user engagement surfaces to enhance user security.
- Enhancing ML-based Account Takeover (ATO) protection measures for TikTok's US customer base to reduce political account and high value account ATO prevalence by **40%** during crucial 2024 election year.
- **Recognition:** Two high-performance bonuses (Exceeded Expectations - E rating) in three years due to company-wide impact. TikTok Hackathon winner for a productionalized RiskAI Copilot.

DataVisor

Mountain View, CA

Senior Data Scientist, Fraud Detection & Research

August 2020 – December 2023 (3 years 5 months)

- Utilizing open source and proprietary machine learning techniques to solve financial fraud problems such as payments fraud, application fraud, account takeover and money laundering with less than **1% false positive rate** and over **70% recall**.
- Led multiple projects and mentored junior data scientists, applying graph analytics to semi-labeled machine learning problems and building identity graph solutions for entity resolution on massive datasets.
- Investigated fraud patterns and developed techniques to boost the performance of DataVisor's proprietary unsupervised machine learning models for some of the largest financial and eCommerce platforms in the world.
- Deploying production models, designing and implementing story-telling dashboards to monitor fraud program performance, and accelerating development of governance documentation by **30%**.
- Supporting go-to-market initiatives by improving deal conversion by **50%**, resulting in closing over **\$1.2 million** in new business.

PwC

New York, NY

Data Scientist (Fraud Detection) - Senior Associate

February 2018 – August 2020 (2 years 7 months)

- Driving cutting edge data analytics using big data tools and machine learning to build, monitor and enhance fraud strategies at financial services clients.
- Leading a team of backend, UI and machine learning engineers from the US, Europe, India and China to build PwC's fraud risk advanced analytics integrated solution.
- Building strategic campaigns around acquisition, marketing, lifecycle, fraud and operations optimization for the customer base for financial services clients using analytics driven decision making, helping deliver top line savings of over **\$10MN**.

EARLY EXPERIENCE

PwC | Data Scientist (Analytics) Intern

Philadelphia, PA | June 2017 – August 2017

- Analyzed fraud in the Forensics Technology Services (FTS) practice; leveraged SQL and Tableau to improve investigation precision by **15%**.

Scratchwork Tool | Software Developer - Senior Design

Princeton, NJ (Remote) | August 2017 – December 2017

- Project lead for a team of six students building the iOS (Swift) application for the Scratchwork handwriting tool.

Penn State University | Learning Assistant (CMPSC 465)

State College, PA | August 2017 – December 2017

- Held weekly data structures and algorithms office hours for **60+ students** and created a community for **competitive programming practice**.

Oracle | Software Engineering Co-op

Conshohocken, PA | June 2016 – December 2016

- Automated testing infrastructure for Primavera Prime using Integration and Selenium suites, improving bug capture rate by over **50%**.

SKILLS

Machine Learning Sequence Models · Gradient-Boosted Trees (CatBoost) · Graph Analytics & Entity Resolution · Unsupervised Detection · LLM Applications

Programming Python · SQL · PyTorch · TensorFlow · Spark · Java · Golang · C · NetworkX · JavaScript · NodeJS · HTML · Swift

Packages and Tools Spark · MLlib · Hadoop · Scikit Learn · PostgreSQL · Tableau · PowerBI · Alteryx · AWS EC2 · AWS EMR · GCP Kubernetes

EDUCATION

Georgia Institute of Technology

College of Computing (OMSCS)

MS in Computer Science (Machine Learning)

Grad. May 2023

Coursework Machine Learning · Deep Learning · Reinforcement Learning · Artificial Intelligence · Machine Learning for Trading · Data and Visual Analytics · Graduate Algorithms · Database Systems · Software Development Process

Penn State University

College of Engineering | Univ. Park, PA

BS in Computer Engineering

Grad. Dec 2017

Cum Laude | Dean's List (All Semesters) | Cum. GPA: 3.84 / 4.0

RESEARCH & HONORS

Research Open ROV Underwater Robot ([Robert Avanzato](#)) · Backward heat equation ([Dr. Matthew Fury](#)) · C++ Teaching Tool (Dr. Tom Warms)

Awards TikTok Hackathon Winner (2024) · Dr. Ayoub Mathematics Award (2015) · President's Freshman Award (2014)

Societies Tau Beta Pi Honor Society · HackPSU · TedxPSU · Vice President, Robotics Club